Intelligent Medical Report Analysis System

Milestone 5 — Evaluation of Model Performance & Analysis of Results

Project Domain: DL / GenAI / NLP

DSAI LAB • Group 2

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System Pipeline & Evaluation Scope

4-Stage Pipeline Under Evaluation

1 **PaddleOCR (PP-OCRv4)**
Perception & OCR: text bounds, tables, CLAHE + rotation correction, <0.6 confidence drop

2 **ClinicalBERT (NER)**
BIO token tagging (O, B/I-TEST, B-VALUE, B-UNIT) + fuzzy match vs 649 MIMIC labels

3 **Normalization → JSON**
Structured key–values: test, value, unit, reference range, status flags

4 **BioMistral 7B (QLoRA)**
Patient-friendly risk explanation & clinical recommendations

What Milestone 5 Measures

Extraction engine (ClinicalBERT)
Token & entity-level accuracy on held-out MIMIC-III / MTSamples reports.

Real-world image robustness
End-to-end detection, value & unit accuracy across 20 diverse lab-report images.

Generative summarizer (BioMistral 7B)
NLL / perplexity, ROUGE-L, BERTScore & readability of patient-facing text.

Evaluation Datasets & Environment

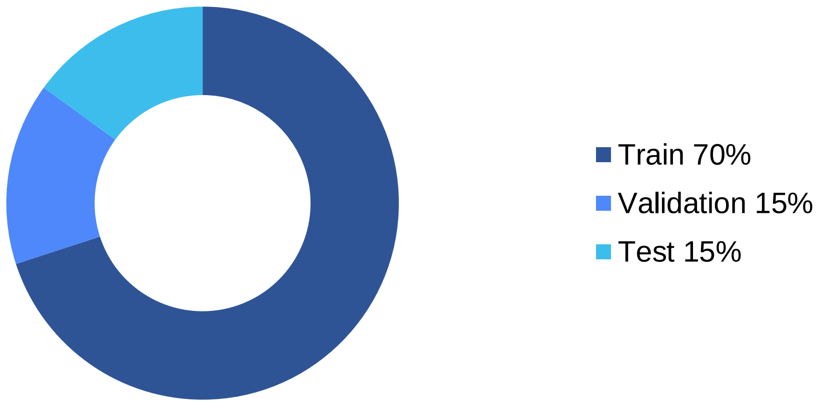
Three Evaluation Sets

Evaluation Set	Size	Split
ClinicalBERT (NER)	21,628 reports	70 / 15 / 15
Real-world lab images	20 images	6 types
BioMistral 7B (LLM)	507 instructions	324 / 81 / 102

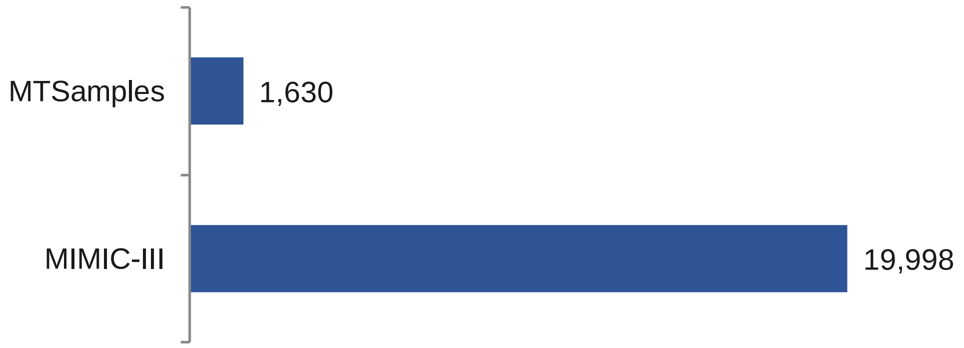
Reproducibility Environment

Component	Specification
GPU	2 × NVIDIA Tesla T4 (16 GB each)
Frameworks	PyTorch 2.2+, HF transformers, peft, bitsandbytes
OCR	PaddleOCR (PP-OCRv4)
Eval tooling	seqeval, Optuna, scikit-learn
Seed	42 (splits + adapter init)

ClinicalBERT Split (21,628)



Source Distribution



ClinicalBERT Extraction Performance

99.97%	99.97%	0.00239	373,556
Weighted F1	Token Accuracy	Test Loss	Tokens Evaluated

Targets vs Achieved (Test Set, 3,245 samples)

Metric	Achieved	Target	Status
Overall Token Accuracy	99.97%	> 95.0%	Exceeded
Entity-Level Macro F1	99.96%	> 90.0%	Exceeded
Overall Weighted F1	99.97%	> 90.0%	Exceeded
Test Loss	0.00239	< 0.01	Met

Confusion Analysis (NER)

Errors are rare and almost entirely O ↔ entity confusion from OCR noise.

Most frequent: O→B-VALUE (23), O→I-TEST (21), O→B-TEST (17), O→B-UNIT (13).

Best checkpoint restored from Epoch 9 (val loss 0.0016) via load_best_model_at_end; F1 saturated at 99.9723%.

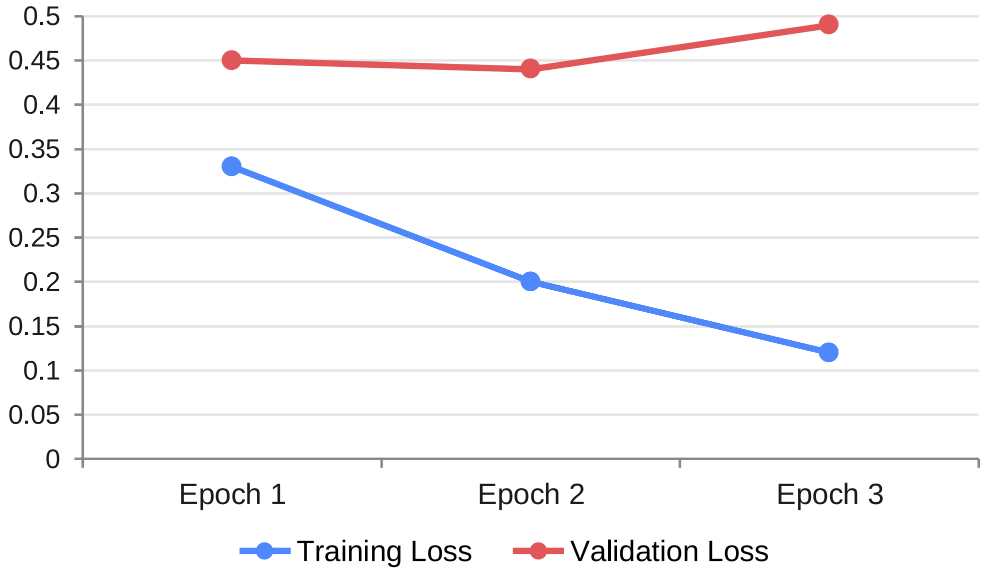
Metrics saturate near ceiling → Optuna search omitted for ClinicalBERT (no headroom; ~84 GPU-hrs for negligible gain).

BioMistral 7B — Generative Performance

Final Metrics (Optuna Trial 5 · r=8, α=16, LR=3.31e-4)

Metric	Value	Interpretation
Eval Loss (NLL)	0.6906	Lowest of all 8 trials
Perplexity	1.99	High next-token certainty
Eval Token Accuracy	80.75%	Agreement with reference
Train→Eval Gap	0.0587	Strong generalisation
ROUGE-L	0.642	Covers reference content
Numeric Fidelity	100%	Every value/unit reproduced exactly
Trainable Params	20.97M	0.2888% of 7.26B

Training vs Validation Loss

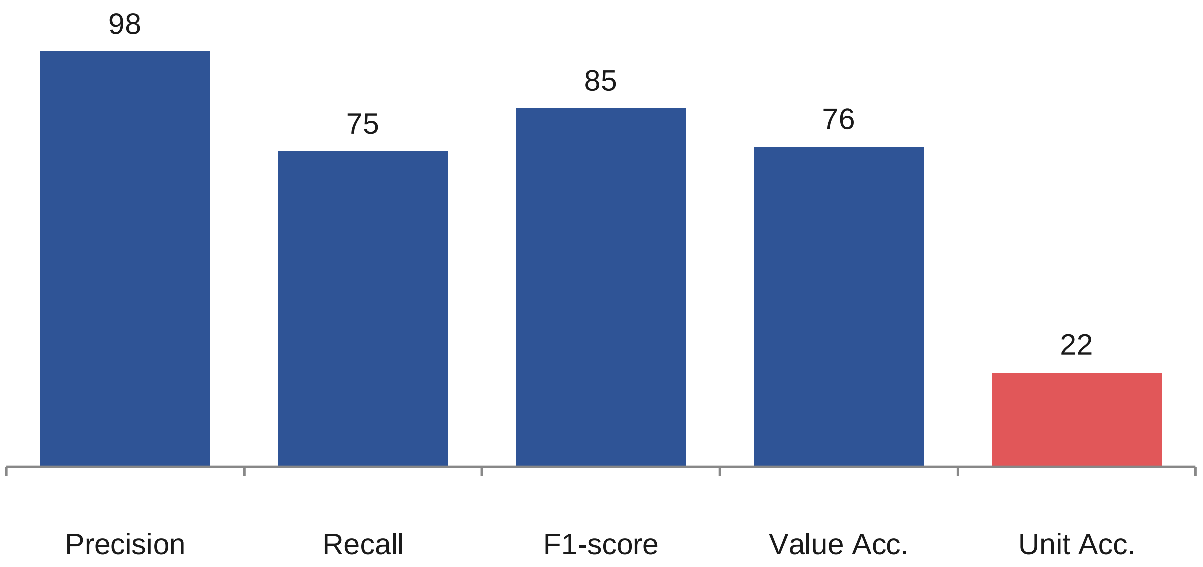


Checkpoint selected at Epoch 2 (val-loss minimum) — textbook small-data overfitting past that point.

Best config modifies only 0.29% of weights yet reaches perplexity 1.99 — efficient QLoRA fine-tuning; the binding constraint is dataset size, not model capacity.

Real-World Image Evaluation

Aggregate Entity Extraction (20 reports · 345 tests)



Unit accuracy (22.3%) is the pipeline's clear bottleneck — the number is right but the unit is wrong or missing.

Performance by Image Type

Image Type	n	Det%	Val%	Unit%
Screenshot	3	100	100	0
Low-Quality	3	84	92	29
Electrolytes	1	71	100	40
Human Captured	2	69	69	45
PDF Report	1	68	68	47
Rotated	4	65	65	13
Standard Lab	6	63	63	30

Screenshots lead — born-digital, no rotation needed. Rotated images still lag post-correction (65% detection, 13% unit): position recovers, reading order doesn't.

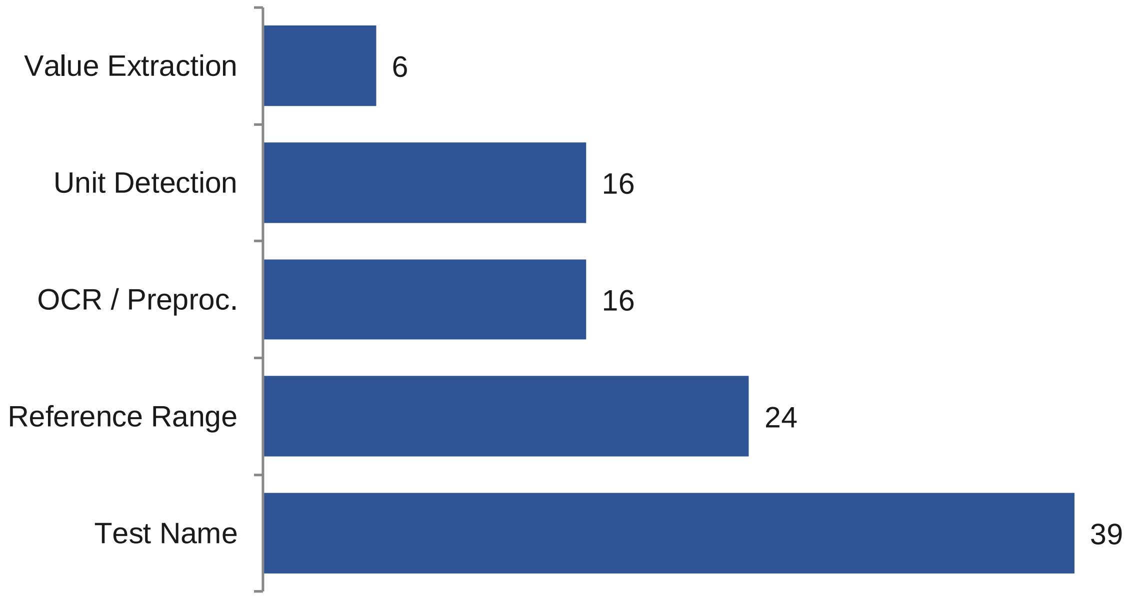
End-to-end success (DARSS composite): 60% Overall Report Success — 12 Successful (≥70%), 3 Partial (60–70%), 5 Unsuccessful (<60%). Rotation correction: 100% (4/4).

Comprehensive Error Analysis

15.7%
OCR & Preprocessing

84.3%
Token Tagging / NER

Error Distribution by Category



Generative Synthesis Errors

Error Type	Frequency
Missed abnormality label	151 / 266 (56.8%)
Missing lab test in summary	88 / 345 (25.5%)
Missing unit in explanation	36 / 266 (13.5%)
Corrupted / false test names	5 false positives
Hallucinated (non-lab tokens)	4 instances

Dataset Bottlenecks

- Limited report-layout diversity (few lab templates)
- OCR errors propagate directly into BIO labels
- Imbalanced O-tag dominance; scarce real camera images

Case Studies & Key Observations

Qualitative Cases

PASS

Successful case
hemoglobin 14.6 · cholesterol 215 → correct NORMAL/HIGH flags; accurate patient-facing narrative, no hallucination.

RECOVER

Rotated 90° image
Garbage raw OCR → after rotation correction all Na/K/Cl tests detected correctly.

FAIL

NER edge cases
Reference-range bound tagged as B-VALUE; merged '25010^9/L' as single value token.

Key Observations & Limitations

- Parameter efficiency: 20.97M params (0.29%) reach eval loss 0.6906, perplexity 1.99.
- Learning-rate dominance: LR drives eval loss more than adapter size; r=8 generalises better than r=64.
- Stability: 99.96%+ F1 across all splits, no leakage or overfitting.
- Readability shortfall: summaries mean grade 10.5 vs grade-8 target — inherited clinical phrasing (a real limitation).
- Constraints: no handwritten-report support; T4 hardware limits.

Interactive Multi-Turn Chat

A stateful wrapper exposes the fine-tuned BioMistral 7B as a conversational assistant: a lab panel opens the session, then free-text follow-ups are answered in context.

How It Works

Stateful history
Full conversation passed on every call, so follow-ups resolve against earlier results; a reset flag starts a fresh case.

Deterministic decoding
Greedy generation (do_sample=False) is reproducible and auditable — important in a medical setting.

Prompt-stripped output
Only newly generated tokens are decoded; runs under torch.no_grad() in a simple REPL until 'exit'.

Observed Behaviour

- **Context retention**
Follow-ups answered against the original result, not in isolation.
- **Format consistency**
Two-part explanation + Recommendation block persists into free chat.
- **Appropriate safety behaviour**
Defers to physicians, declines to diagnose, advises against stopping prescribed medication.

Limitations: greedy decoding causes near-verbatim repetition of the Recommendation block; risk framing is occasionally too strong (“Yes, be concerned”); conversation history grows unbounded toward the context limit.

THANK YOU

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